

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412392
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Colamatteo; Valerio et al.

Sports neural network codec

Abstract

A computing system receives a broadcast video stream of a game. A codec module of the computing system extracts image level features from the broadcast video stream. The codec module includes an object detection portion configured to detect players in the broadcast video stream and a subnet portion attached to the object detection portion. The subnet portion is configured to identify foreground information of the detected players. The codec module provides the image level features to a plurality of task specific modules for analysis. The plurality of task specific modules generates a plurality of outputs based on the image level features.

Inventors: Colamatteo; Valerio (Milan, IT), Evi-Parker; Christopher (London, GB), Padagadi; Sateesh (London, GB), Lucey; Patrick Joseph (Chicago, IL)

Applicant: STATS LLC (Chicago, IL)

Family ID: 86158732

Assignee: STATS LLC (Chicago, IL)

Appl. No.: 18/050331

Filed: October 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230148112 A1	May. 11, 2023

Related U.S. Application Data

us-provisional-application US 63263189 20211028

Publication Classification

Int. Cl.: G06V10/20 (20220101); G06V10/44 (20220101); G06V10/778 (20220101); G06V10/82 (20220101); G06V20/40 (20220101); G06V40/10 (20220101); H04N21/234 (20110101)

U.S. Cl.:

CPC G06V20/42 (20220101); G06V10/778 (20220101);

Field of Classification Search

CPC: G06V (20/42); G06V (10/778); G06V (10/255); G06V (10/454); G06V (10/82); H04N (19/70); H04N (19/176); H04N (19/172); H04N (19/597); H04N (19/46); H04N (21/8456); G06N (3/045); G06N (3/08); G06T (2207/10016)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2008/0192116	12/2007	Tamir et al.	N/A	N/A
2011/0013836	12/2010	Gefen et al.	N/A	N/A
2011/0268320	12/2010	Huang et al.	N/A	N/A
2016/0292510	12/2015	Han et al.	N/A	N/A
2017/0344884	12/2016	Lin et al.	N/A	N/A
2019/0221001	12/2018	Dassa	N/A	G06N 3/045
2020/0234051	12/2019	Lee	N/A	G06V 10/764
2020/0394413	12/2019	Bhanu et al.	N/A	N/A
2022/0292311	12/2021	Song	N/A	G06V 10/774
2024/0161461	12/2023	Zu	N/A	G06V 10/82

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111951249	12/2019	CN	N/A
112115914	12/2019	CN	G06K 9/00664
WO-2020237215	12/2019	WO	G06F 18/2155
2021016901	12/2020	WO	N/A

OTHER PUBLICATIONS

PCT International Application No. PCT/US22/78794, International Search Report and Written Opinion of the International Searching Authority, dated Feb. 2, 2023, 11 pages. cited by applicant
He Kaiming et al: “Mask R-CNN”, Jan. 24, 2018 (Jan. 24, 2018), XP055930853, Retrieved from the Internet:URL: <https://arxiv.org/pdf/1703.06870.pdf> [retrieved on Jun. 14, 2022]. cited by applicant
Jacek Komorowski et al: “FootAndBall: Integraed player and ball detector”, ARXIV.ORG, Cornell

University Library, 201 Olin Library Cornell University Ithaca, NY 14853, Oct. 26, 2020 (Oct. 26, 2020), XP081799873. cited by applicant

Pobar Miran et al: "Mask R-CNN and Optical Flow Based Method for Detection and Marketing of Handball Actions", 2018 11TH International Congress on Image and Signal Processing, Biomedical Engineering and Informatics (CISP-BMEI), IEEE, Oct. 2018 (Oct. 13, 2018), pp. 1-6, XP033511738. cited by applicant

Roman Voeikov et al: "TTNet: Real-time temporal and spatial video analysis of table tennis", ARXIV.ORG, Cornell University Library, 201 Olin Library Cornell University Ithaca, NY 14853, Apr. 21, 2020 (Apr. 21, 2020), XP081649342. cited by applicant

Sah Melike et al: "Review and evaluation of player detection methods in field sports: Comparing conventional and deep learning based methods", Multimedia Tools and Applications., [Online] vol. 82, No. 9, Jun. 3, 2021 (Jun. 3, 2021), pp. 13141-13165, XP093265850, US ISSN: 1380-7501, DOI: 10.1007/s11042-021-11071-z Retrieved from the Internet:URL:<https://link.springer.com/article/10.1007/s11042-021-11071-z/fulltext.html>> [retrieved on Apr. 2, 2025]. cited by applicant

Zhang Yiqing et al: "Mask-Refined R-CNN: A Network for Refining Object Details in Instance Segmentation", Sensors, vol. 20, No. 4, Feb. 13, 2020 (Feb. 13, 2020), p. 1010, XP093266346. cited by applicant

Primary Examiner: Villecco; John

Assistant Examiner: Allen; Kyla Guan-Ping Tiao

Attorney, Agent or Firm: Bookoff McAndrews PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application Ser. No. 63/263,189, filed Oct. 28, 2021, which is hereby incorporated by reference in its entirety.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to sports neural network encoder for sporting contests.

BACKGROUND

(2) Increasingly, users are opting to forego a traditional cable subscription in favor of one of the various streaming services readily available today. With this shift, leagues across a variety of sports have become more interested in contracting with one of these streaming services for providing their content to end users.

SUMMARY

(3) In some embodiments, a method is disclosed herein. A computing system receives a broadcast video stream of a game. A codec module of the computing system extracts image level features from the broadcast video stream. The codec module includes an object detection portion configured to detect players in the broadcast video stream and a subnet portion attached to the object detection portion. The subnet portion is configured to identify foreground information of the detected players. The codec module provides the image level features to a plurality of task specific modules for analysis. The plurality of task specific modules generates a plurality of outputs based on the image level features.

(4) In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium includes one or more sequences of instructions, which,

when executed by a processor, causes a computing system to perform operations. The operations include receiving, by the computing system, a broadcast video stream of a game. The operations further include extracting, via a codec module of the computing system, image level features from the broadcast video stream. The codec module includes an object detection portion configured to detect players in the broadcast video stream and a subnet portion attached to the object detection portion. The subnet portion is configured to identify foreground information of the detected players. The operations further include providing, by the codec module, the image level features to a plurality of task specific modules for analysis. The operations further include generating, by the plurality of task specific modules, a plurality of outputs based on the image level features.

(5) In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory has programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations. The operations include receiving a broadcast video stream of a game. The operations further include extracting, via a codec module, image level features from the broadcast video stream. The codec module includes an object detection portion configured to detect players in the broadcast video stream and a subnet portion attached to the object detection portion. The subnet portion is configured to identify foreground information of the detected players. The operations further include providing, by the codec module, the image level features to a plurality of task specific modules for analysis. The operations further include generating, by the plurality of task specific modules, a plurality of outputs based on the image level features.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrated only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a block diagram illustrating a computing environment, according to example embodiments.

(3) FIG. 2 is a block diagram that illustrates exemplary components of computing system, according to example embodiments.

(4) FIG. 3 is a block diagram that illustrates a machine learning architecture implemented by codec module, according to example embodiments.

(5) FIG. 4 is a flow diagram illustrating a method of processing a broadcast video feed, according to example embodiments.

(6) FIG. 5A is a block diagram illustrating a computing device, according to example embodiments.

(7) FIG. 5B is a block diagram illustrating a computing device, according to example embodiments.

(8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(9) The efficient extraction of human understandable data in sports vision analysis is typically a highly computational process based on the accomplishment of multiple tasks through stand-alone

designs and developed modules. Conventionally, these modules are typically sequentially stacked for producing the desired output (e.g., player position, court geometry, etc.). This working schema is vertically structured and, thus, computationally highly redundant because each module independently encodes and decodes information from a single visual input.

(10) Further, conventional approaches to object detection are unable to also support the identification of foreground information of the objects. Conventionally, operators had to employ two separate models: a first model configured to detect objects; and a second model configured to identify foreground information of the objects. In the context of real-time applications, such as in detecting players in sports, such two-step approach is time consuming and cannot support real-time functionality.

(11) To improve upon conventional processes, one or more techniques provided herein provide a universal approach for unifying many of sports' visual information extraction tasks into a single framework. Such functionality may be accomplished by attaching a mask subnet to an object detection module. This approach allows for object detection and foreground identification using a single machine learning architecture. In this manner, the architecture disclosed herein can be efficiently deployed in real-time applications.

(12) FIG. 1 is a block diagram illustrating a computing environment **100**, according to example embodiments. Computing environment **100** may include tracking system **102**, organization computing system **104**, and one or more client devices **108** communicating via network **105**.

(13) Network **105** may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network **105** may connect terminals, services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connection be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

(14) Network **105** may include any type of computer networking arrangement used to exchange data or information. For example, network **105** may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment **100** to send and receive information between the components of environment **100**.

(15) Tracking system **102** may be positioned in a venue **106**. For example, venue **106** may be configured to host a sporting event that includes one or more agents **112**. Tracking system **102** may be configured to capture the motions of all agents (i.e., players) on the playing surface, as well as one or more other objects of relevance (e.g., ball, referees, etc.). In some embodiments, tracking system **102** may be an optically-based system using, for example, a plurality of fixed cameras. For example, a system of six stationary, calibrated cameras, which project the three-dimensional locations of players and the ball onto a two-dimensional overhead view of the court may be used. In another example, a mix of stationary and non-stationary cameras may be used to capture motions of all agents on the playing surface as well as one or more objects or relevance. As those skilled in the art recognize, utilization of such tracking system (e.g., tracking system **102**) may result in many different camera views of the court (e.g., high sideline view, free-throw line view, huddle view, face-off view, end zone view, etc.). In some embodiments, tracking system **102** may be used for a broadcast feed of a given match. In such embodiments, each frame of the broadcast feed may be stored in a game file **110**.

(16) In some embodiments, game file **110** may further be augmented with other event information corresponding to event data, such as, but not limited to, game event information (pass, made shot, turnover, etc.) and context information (current score, time remaining, etc.).

(17) Tracking system **102** may be configured to communicate with organization computing system **104** via network **105**. For example, tracking system **102** may be configured to provide organization computing system **104** with a broadcast stream of a game or event in real-time or near real-time via network **105**.

(18) Organization computing system **104** may be configured to process the broadcast stream of the game and provide various insights or metrics related to the game to client devices **108**.

Organization computing system **104** may include at least a web client application server **114**, a pre-processing agent **116**, data store **118**, codec module **120**, and task specific modules **122**. Each of pre-processing agent **116**, codec module **120**, and task specific modules **122** may be comprised of one or more software modules. The one or more software modules may be collections of code or instructions stored on a media (e.g., memory of organization computing system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that is interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather as a result of the instructions.

(19) Data store **118** may be configured to store one or more game files **124**. Each game file **124** may include video data of a given match. For example, the video data may correspond to a plurality of video frames captured by tracking system **102**. In some embodiments, the video data may correspond to broadcast data of a given match, in which case, the video data may correspond to a plurality of video frames of the broadcast feed of a given match.

(20) Pre-processing agent **116** may be configured to process data retrieved from data store **118**. For example, pre-processing agent **116** may be configured to generate game files **124** stored in data store **118**. For example, pre-processing agent **116** may be configured to generate a game file **124** based on data captured by tracking system **102**. In some embodiments, pre-processing agent **116** may further be configured to store tracking data associated with each game in a respective game file **124**. Tracking data may refer to the (x, y) coordinates of all players and balls on the playing surface during the game. In some embodiments, pre-processing agent **116** may receive tracking data directly from tracking system **102**. In some embodiments, pre-processing agent **116** may derive tracking data from the broadcast feed of the game.

(21) Codec module **120** may be configured to process broadcast video data received by organization computing system **104**. In some embodiments, codec module **120** may process broadcast video data in real-time or near-real time. Codec module **120** may be representative of a neural network architecture configured to extract a plurality of features from the broadcast video data for downstream analysis by task specific modules **122**. Codec module **120** may be configured to generate input serving multiple task specific modules **122**. Such architecture may allow codec module **120** to function as a generalized sports image encoder. Exemplary features that may be extracted may include, but are not limited to, player detection during the game, discerning players from spectators, playing ball detection, team identification related to any player on the playing surface, jersey numbers optical detection and recognition, player re-identification by appearance, instance segmentation, score board detection, and the like.

(22) Codec module **120** may successively refine one or more encodings (which may include the embeddings) of the input visual data by distributing the encodings to several heads of the neural network architecture for single task specialization. This multiplicity of sports-encoding heads with a single features' extraction moment allows for reuse of backbone encodings in a runtime efficient manner due to the parallelism. As such, codec module **120** may be suitable for both on-line and off-line analysis.

(23) Task specific modules **122** may be representative of various prediction models for generating

insights or statistics related to events within the broadcast video data feed. In some embodiments, task specific modules **122** may receive output from codec module **120** for generating downstream predictions. For example, task specific modules **122** may be provided with various features extracted from the broadcast video data feed from codec modules **120**. Exemplary features may include, but are not limited to, foreground pixel locations and player location information.

(24) Client device **108** may be in communication with organization computing system **104** via network **105**. Client device **108** may be operated by a user. For example, client device **108** may be a mobile device, a tablet, a desktop computer, a set-top box, a streaming player, or any computing system capable of receiving, rendering, and presenting video data to the user. Users may include, but are not limited to, individuals such as, for example, subscribers, clients, prospective clients, or customers of an entity associated with organization computing system **104**, such as individuals who have obtained, will obtain, or may obtain a product, service, or consultation from an entity associated with organization computing system **104**.

(25) Client device **108** may include at least application **126**. Application **128** may be representative of a web browser that allows access to a website or a stand-alone application. Client device **108** may access application **126** to access one or more functionalities of organization computing system **104**. Client device **108** may communicate over network **105** to request a webpage, for example, from web client application server **114** of organization computing system **104**. For example, client device **108** may be configured to execute application **126** to access one or more insights or statistics generated by task specific modules **122**. The content that is displayed to client device **108** may be transmitted from web client application server **114** to client device **108**, and subsequently processed by application **126** for display through a graphical user interface (GUI) of client device **108**.

(26) FIG. 2 is a block diagram that illustrates exemplary components of computing environment **100**, according to example embodiments. As shown, a broadcast video stream **202** may be provided to codec module **120**. Codec module **120** may be configured to extract features **204** from the broadcast video feed. Exemplary features **204** may include, but are not limited to player detection during the game, discerning players from spectators, playing ball detection, team identification related to any player on the playing surface, jersey numbers optical detection and recognition, player re-identification by appearance, instance segmentation, score board detection, and the like. Features **204** may be provided by codec module **120** to task specific modules **122** for downstream processing. For example, task specific modules **122** may utilize features **204** to generate various insights or statistics (e.g., output **206**) related to events in the broadcast video stream. In this manner, codec module **120** may only need to process the broadcast video feed once and pass those extracted features to task specific modules **122**.

(27) FIG. 3 is a block diagram that illustrates a machine learning architecture **300** implemented by codec module **120**, according to example embodiments.

(28) As shown, machine learning architecture **300** may include an object detection portion **302** with an attached subnet portion **304**. Object detection portion **302** may be trained to identify objects in a video. For example, object detection portion **302** may be trained to identify players in a broadcast video stream. In some embodiments, object detection portion **302** may be representative of an object detection architecture, such as, but not limited to, a YOLOV5 architecture. YOLOv5 architecture is an object detection algorithm that is configured to divide images into a grid system, with each grid responsible for detecting objects within itself.

(29) As shown, object detection portion **302** may include a backbone **306**, a neck **308**, and a head **310**. Backbone **306** may be configured to extract image level features from the video. In some embodiments, backbone **306** may be representative of a convolutional neural network architecture. For example, as shown, backbone **306** may include several convolutional layers configured to extract the image features. Backbone **306** may provide extracted image level features to neck **308**. Neck **308** may be configured to aggregate the extracted image level features. For example, neck **308** may be configured to collect image level features from a plurality of different levels. In some

embodiments, the output generated by neck **308** may be representative of floating point values that indicate a likely position of objects or players in the video. Head **310** may be configured to identify a location of objects in the video based on input from neck **308**. For example, head **310** may include a plurality of convolutions. Each convolution may be configured to use different resolutions to extract image features to detect player location in the video. In this manner, head **310** may increase or improve the stability of detection across different environments. Accordingly, in some embodiments, as output, object detection portion **302** may provide player locations in the video.

(30) In some embodiments, output from each convolutional may be provided to a non-maximum suppression (NMS) function **330**. NMS function **330** may be configured to take each bounding box coordinate generated by the plurality of convolutions for a given player and combine them into a single bounding box identifying a location of the player.

(31) Subnet portion **304** may be attached to object detection portion **302**. For example, as shown, subnet portion **304** may be attached to object detection portion **302** to the output of neck **308**. Accordingly, in this manner, subnet portion **304** may receive, as input, the direct output from neck **308** as well as the output generated from NMS function **330**.

(32) Subnet portion **304** may include a plurality of operators **312** and a plurality of mask subnets **314**. In some embodiments, each operator of plurality of operators **312** may be representative of a region of interest align (RoIAlign) operation. Output from plurality of operators **312** may be provided to a respective mask subnet **314**. Mask subnet **314** may be configured to generate pixel level information to detect the foreground information of each player. In some embodiments, mask subnet **314** may use thresholding to generate a player mask.

(33) In this manner, machine learning architecture **300** is able to detect player locations in a video feed and generate foreground information that may be used for downstream processes using a single model.

(34) In some embodiments, training machine learning architecture **300** to detect player locations and generate foreground information may be done in a two-step process. For example, in some embodiments, object detection portion **302** may be first trained independent of subnet portion **304**. In this manner, object detection portion **302** may achieve a threshold level of accuracy for detecting player locations in the video feed. Following training of object detection portion **302**, subnet portion **304** may be attached to neck **308** for further training. In some embodiments, the initial weights of machine learning architecture **300** with subnet portion **304** attached to object detection portion **302** may be set to the final weights generated during independent training of object detection portion **302**.

(35) FIG. 4 is a flow diagram illustrating a method **400** of generating interactive broadcast video data, according to example embodiments. Method **400** may begin at step **402**.

(36) At step **402**, organization computing system **104** may receive a broadcast video stream for a game or event. In some embodiments, broadcast video stream may be provided by tracking system **102**. In some embodiments, the broadcast video stream may be provided in real-time or near real-time.

(37) At step **404**, organization computing system **104** may extract features from the broadcast video stream. For example, codec module **120** may be representative of a neural network backbone configured to analyze and extract a plurality of features from the broadcast video stream. Exemplary features **204** may include, but are not limited to player detection during the game, discerning players from spectators, playing ball detection, team identification related to any player on the playing surface, jersey numbers optical detection and recognition, player re-identification by appearance, instance segmentation, score board detection, and the like.

(38) At block **406**, organization computing system **104** may generate a plurality of artificial intelligence insights or metrics based on the extracted features. For example, codec module **120** may feed or provide input to multiple heads, i.e., task specific modules **122**. Task specific modules

122 may utilize the extracted features to generate the plurality of artificial intelligence insights or metrics. Due to the architecture of codec module **120**, codec module **120** does not need to extract features each time for each task specific module **122**. Instead, codec module **120** may extract the plurality of features in a single pass, and may provide those features to task specific modules **122** for analysis.

(39) At block **408**, organization computing system **104** may provide the artificial intelligence insights or metrics to an end user. For example, organization computing system **104** may provide the artificial intelligence insights or metrics to application **126** executing on client device **108**

(40) FIG. 5A illustrates an architecture of computing system **500**, according to example embodiments. System **500** may be representative of at least a portion of organization computing system **104**. One or more components of system **500** may be in electrical communication with each other using a bus **505**. System **500** may include a processing unit (CPU or processor) **510** and a system bus **505** that couples various system components including the system memory **515**, such as read only memory (ROM) **520** and random access memory (RAM) **525**, to processor **510**. System **500** may include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **510**. System **500** may copy data from memory **515** and/or storage device **530** to cache **512** for quick access by processor **510**. In this way, cache **512** may provide a performance boost that avoids processor **510** delays while waiting for data. These and other modules may control or be configured to control processor **510** to perform various actions. Other system memory **515** may be available for use as well. Memory **515** may include multiple different types of memory with different performance characteristics. Processor **510** may include any general purpose processor and a hardware module or software module, such as service 1 **532**, service 2 **534**, and service 3 **536** stored in storage device **530**, configured to control processor **510** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **510** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(41) To enable user interaction with the computing system **500**, an input device **545** may represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **535** (e.g., display) may also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems may enable a user to provide multiple types of input to communicate with computing system **500**. Communications interface **540** may generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(42) Storage device **530** may be a non-volatile memory and may be a hard disk or other types of computer readable media which may store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **525**, read only memory (ROM) **520**, and hybrids thereof.

(43) Storage device **530** may include services **532**, **534**, and **536** for controlling the processor **510**. Other hardware or software modules are contemplated. Storage device **530** may be connected to system bus **505**. In one aspect, a hardware module that performs a particular function may include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **510**, bus **505**, output device **535**, and so forth, to carry out the function.

(44) FIG. 5B illustrates a computer system **550** having a chipset architecture that may represent at least a portion of organization computing system **104**. Computer system **550** may be an example of computer hardware, software, and firmware that may be used to implement the disclosed technology. System **550** may include a processor **555**, representative of any number of physically

and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. Processor 555 may communicate with a chipset 560 that may control input to and output from processor 555. In this example, chipset 560 outputs information to output 565, such as a display, and may read and write information to storage device 570, which may include magnetic media, and solid-state media, for example. Chipset 560 may also read data from and write data to RAM 575. A bridge 580 for interfacing with a variety of user interface components 585 may be provided for interfacing with chipset 560. Such user interface components 585 may include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system 550 may come from any of a variety of sources, machine generated and/or human generated.

(45) Chipset 560 may also interface with one or more communication interfaces 590 that may have different physical interfaces. Such communication interfaces may include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein may include receiving ordered datasets over the physical interface or be generated by the machine itself by processor 555 analyzing data stored in storage device 570 or RAM 575. Further, the machine may receive inputs from a user through user interface components 585 and execute appropriate functions, such as browsing functions by interpreting these inputs using processor 555.

(46) It may be appreciated that example systems 500 and 550 may have more than one processor 510 or be part of a group or cluster of computing devices networked together to provide greater processing capability.

(47) While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and can be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

(48) It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

Claims

1. A method comprising: receiving, by a computing system, a broadcast video stream of a game; extracting, via a codec module of the computing system, image level features from the broadcast video stream, the codec module comprising an object detection portion that detects players in the broadcast video stream and a subnet portion attached to and downstream of the object detection portion, wherein the subnet portion identifies foreground information of the detected players; providing, by the codec module, the image level features to a plurality of task specific modules for

analysis; and generating, by the plurality of task specific modules, a plurality of outputs based on the image level features.

2. The method of claim 1, wherein the object detection portion comprises: a backbone that extracts image level features from the broadcast video stream; a neck downstream of the backbone, wherein the neck aggregates the extracted image level features; and a head downstream of the neck, wherein the head identifies locations of players in the broadcast video stream based on the extracted image level features.

3. The method of claim 2, wherein the head comprises a plurality of convolutions, wherein each convolution identifies a location of a player at varying resolutions.

4. The method of claim 3, wherein the codec module further comprises: a non-maximum suppression function downstream of the head, wherein the non-maximum suppression function combines the identified locations of the player at varying resolutions to generate a single location for the player.

5. The method of claim 2, wherein the subnet portion is attached to the neck.

6. The method of claim 2, wherein the subnet portion receives input from the neck, wherein the input from the neck is output generated by the neck, the output comprising floating point values indicating a likely position of players in the broadcast video stream.

7. The method of claim 1, further comprising: training, by the computing system, the object detection portion independent of the subnet portion; and after training the object detection portion, training, by the computing system, the object detection portion with the subnet portion attached thereto.

8. A non-transitory computer readable medium comprising one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations comprising: receiving, by the computing system, a broadcast video stream of a game; extracting, via a codec module of the computing system, image level features from the broadcast video stream, the codec module comprising an object detection portion that detects players in the broadcast video stream and a subnet portion attached to and downstream of the object detection portion, wherein the subnet portion identifies foreground information of the detected players; providing, by the codec module, the image level features to a plurality of task specific modules for analysis; and generating, by the plurality of task specific modules, a plurality of outputs based on the image level features.

9. The non-transitory computer readable medium of claim 8, wherein the object detection portion comprises: a backbone that extracts image level features from the broadcast video stream; a neck downstream of the backbone, wherein the neck aggregates the extracted image level features; and a head downstream of the neck, wherein the head identifies locations of players in the broadcast video stream based on the extracted image level features.

10. The non-transitory computer readable medium of claim 9, wherein the head comprises a plurality of convolutions, wherein each convolution identifies a location of a player at varying resolutions.

11. The non-transitory computer readable medium of claim 10, wherein the codec module further comprises: a non-maximum suppression function downstream of the head, wherein the non-maximum suppression function combines the identified locations of the player at varying resolutions to generate a single location for the player.

12. The non-transitory computer readable medium of claim 9, wherein the subnet portion is attached to the neck.

13. The non-transitory computer readable medium of claim 9, wherein the subnet portion receives input from the neck, wherein the input from the neck is output generated by the neck, the output comprising floating point values indicating a likely position of players in the broadcast video stream.

14. The non-transitory computer readable medium of claim 8, further comprising: training, by the

- computing system, the object detection portion independent of the subnet portion; and after training the object detection portion, training, by the computing system, the object detection portion with the subnet portion attached thereto.
15. A system comprising: a processor; and a memory having programming instructions stored thereon, which, when executed by the processor, cause the system to perform operations comprising: receiving a broadcast video stream of a game; extracting, via a codec module, image level features from the broadcast video stream, the codec module comprising an object detection portion that detects players in the broadcast video stream and a subnet portion attached to and downstream of the object detection portion, wherein the subnet portion identifies foreground information of the detected players; providing, by the codec module, the image level features to a plurality of task specific modules for analysis; and generating, by the plurality of task specific modules, a plurality of outputs based on the image level features.
16. The system of claim 15, wherein the object detection portion comprises: a backbone that extracts image level features from the broadcast video stream; a neck downstream of the backbone, wherein the neck aggregates the extracted image level features; and a head downstream of the neck, wherein the head identifies locations of players in the broadcast video stream based on the extracted image level features.
17. The system of claim 16, wherein the head comprises a plurality of convolutions, wherein each convolution identifies a location of a player at varying resolutions.
18. The system of claim 17, wherein the codec module further comprises: a non-maximum suppression function downstream of the head, wherein the non-maximum suppression function combines the identified locations of the player at varying resolutions to generate a single location for the player.
19. The system of claim 16, wherein the subnet portion receives input from the neck, wherein the input from the neck is output generated by the neck, the output comprising floating point values indicating a likely position of players in the broadcast video stream.
20. The system of claim 15, further comprising: training the object detection portion independent of the subnet portion; and after training the object detection portion, training the object detection portion with the subnet portion attached thereto.
-